BASIC ELECTRONICS
(24B11EC111)
UNIT-2
Lecture-8 (Clamper Circuit)

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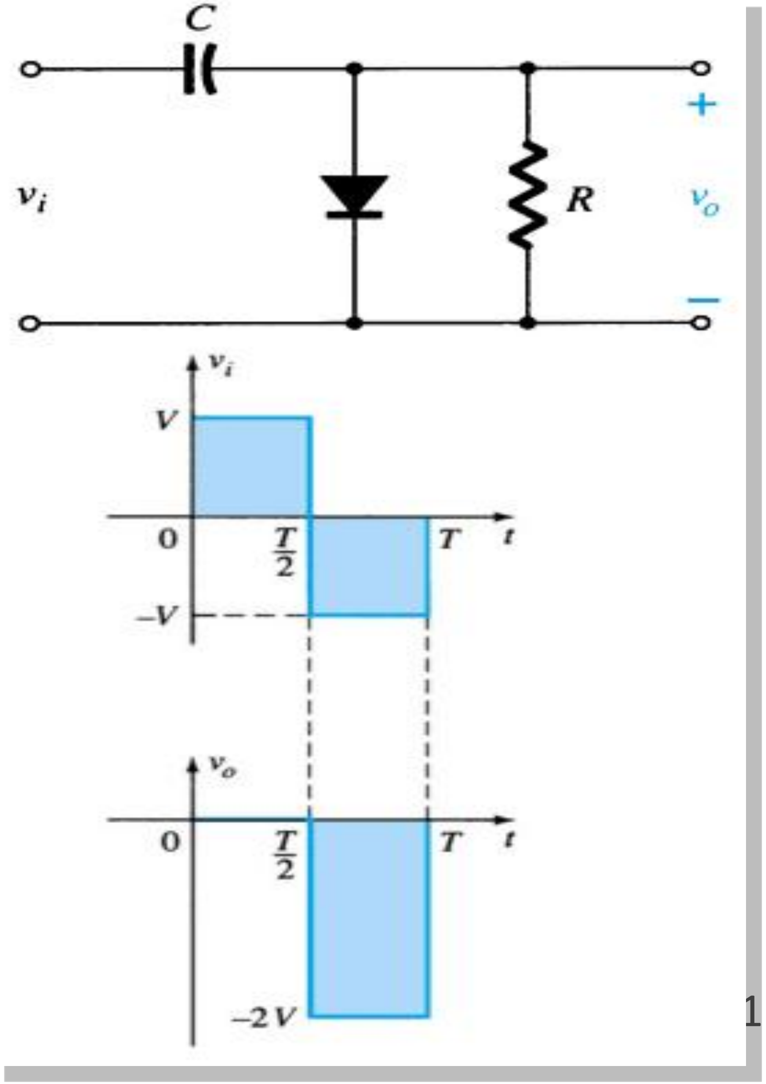
Introduction of clampers

- A clamper is a network constructed of a diode, a resistor, and a capacitor that shifts a waveform to a different dc level without changing the appearance of the applied signal.



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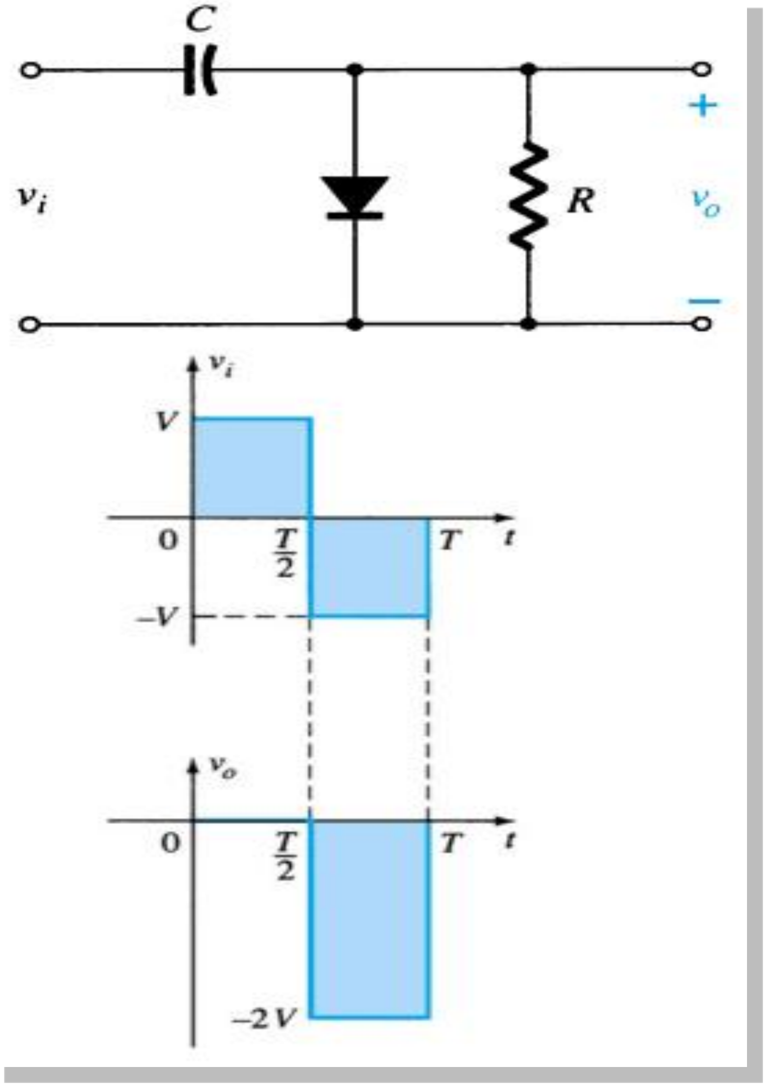
Classifications

- Negative clampers
- Positive clampers
- Biased clampers

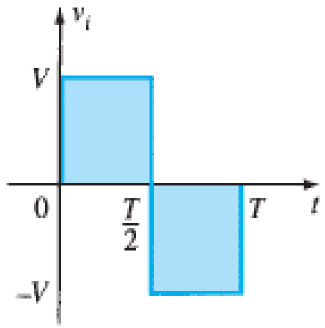
Negative clamper

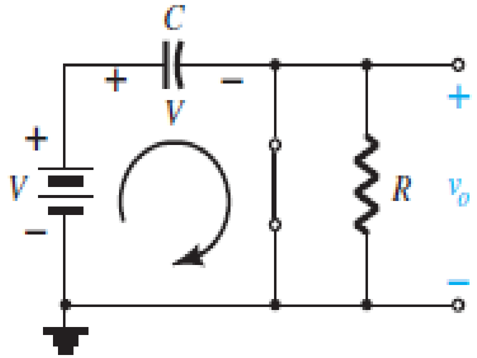
- The Negative Clamping circuit consists of a diode connected in parallel with the load.

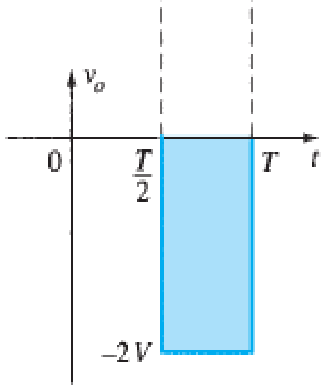
- This type of clamping circuit **shifts the input waveform in a negative direction**, as a result the waveform lies **below a DC reference voltage**.



- During the positive half cycle (interval 0 to T/2)
- $Vo = 0$ V for this time interval.

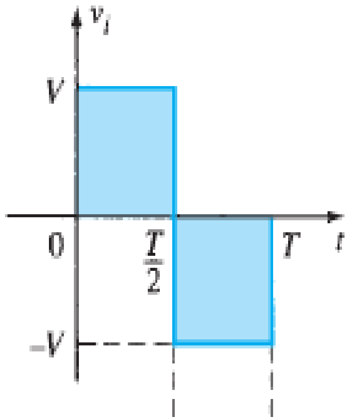


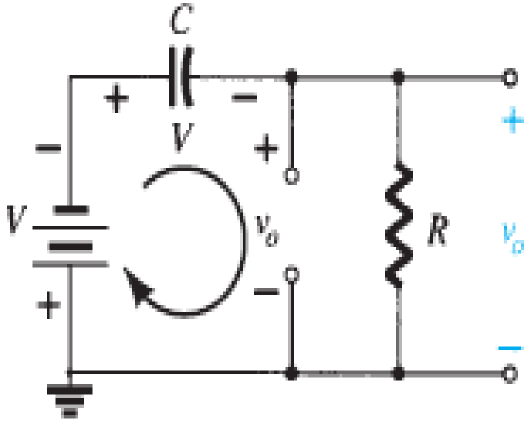


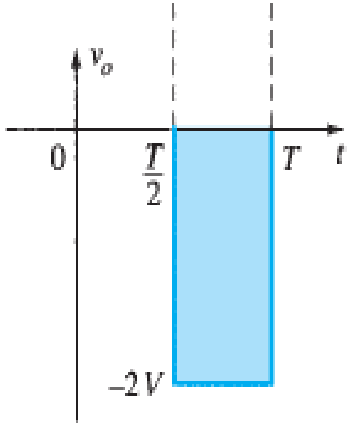


- During this same interval of time, the **time constant** determined by **τ = *RC*** is very small

- During the negative half cycle (interval T/2 to T)
- Applying Kirchhoff’s voltage law around the input loop results in,
- $-V - V - V_0 = 0$
- $V_0 = -2V$

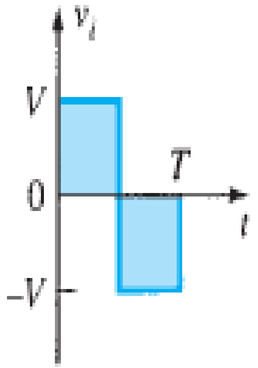


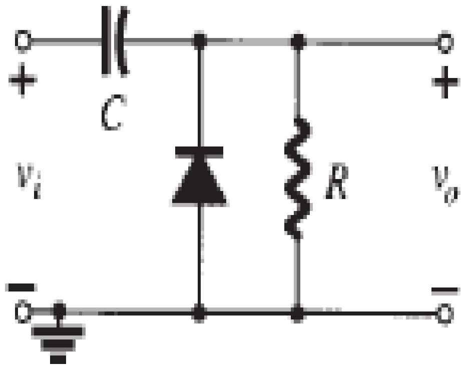


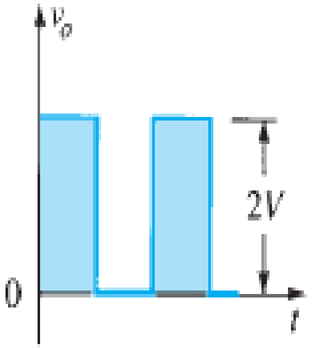


Positive clampers

- This type of clamping circuit shifts the input waveform in a positive direction, as a result the waveform lies above a DC reference voltage.

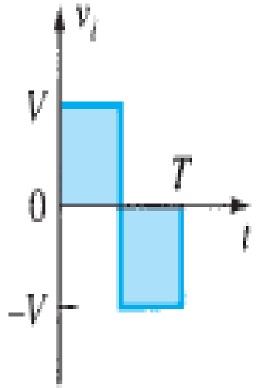


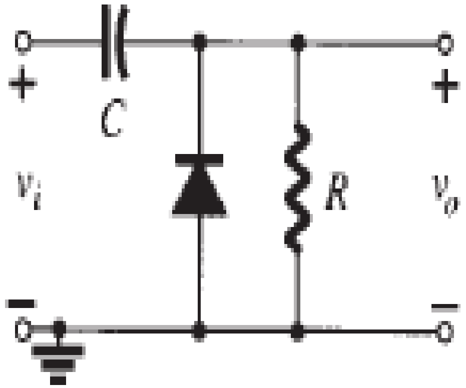


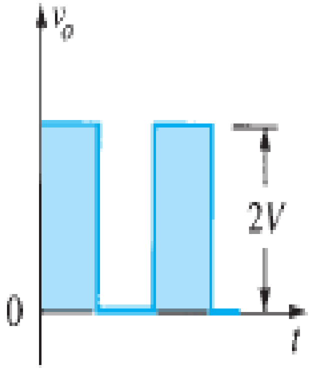


- During the positive half wave cycle → diode goes off
- $V_0 = 2V$

- During the negative half wave cycle →

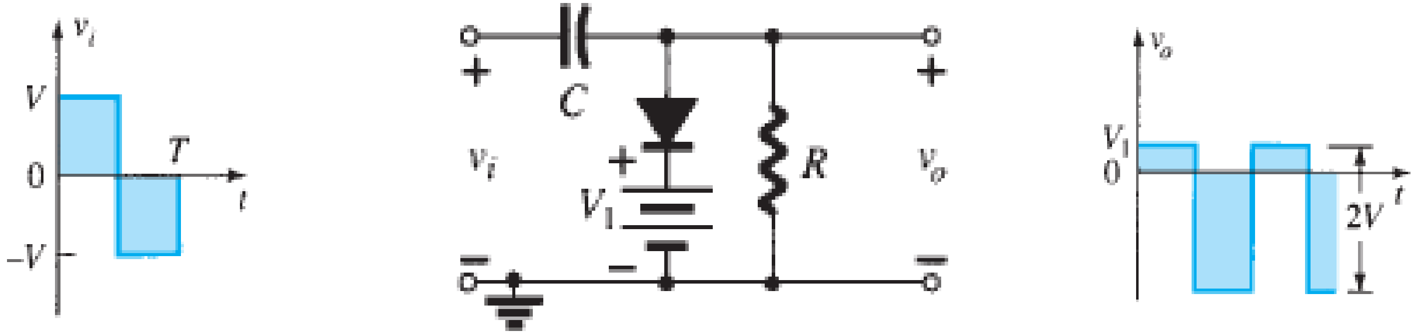






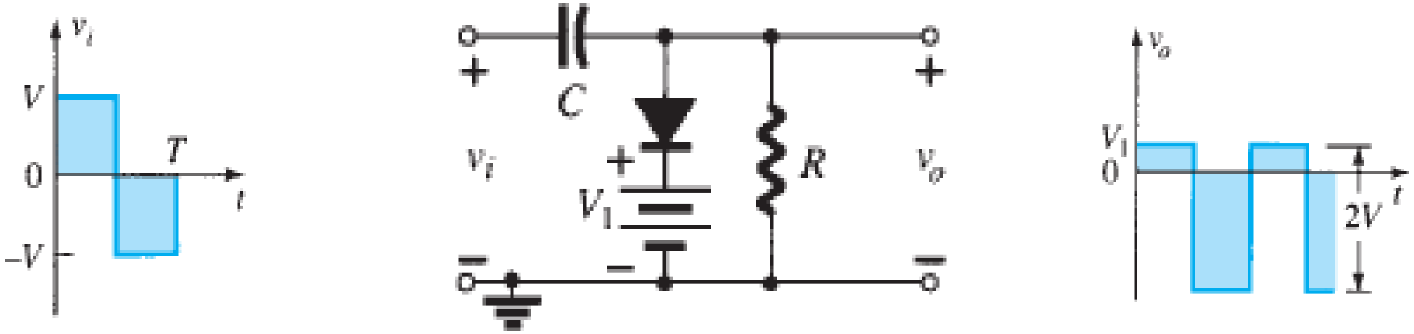
Biased clamper

- The additional dc supply connected within diode as shown in figure.



- For the positive half cycle
- If ,$V_i < V_1$, diode is OFF, $V_0 = Vi - Vc$
- If, $V_i > V_1$, diode is ON, $V_0 = V_1$ and the capacitor charge up to voltage $V_c = Vi - V_1$
- In this case the resistor ***R* is not shorted out by the diode**.

- For the negative half cycle,
- $V_0 = -Vi - Vc \equiv -Vi - Vi + V_1 = -2Vi + V_1$



Summary of clampers

Negative Clamper

Positive Clamper

Negative Clamper with same direction of diode and battery

Positive Clamper with opposite direction of diode and battery

Negative Clamper with opposite direction of diode and battery

Positive Clamper with same direction of diode and battery

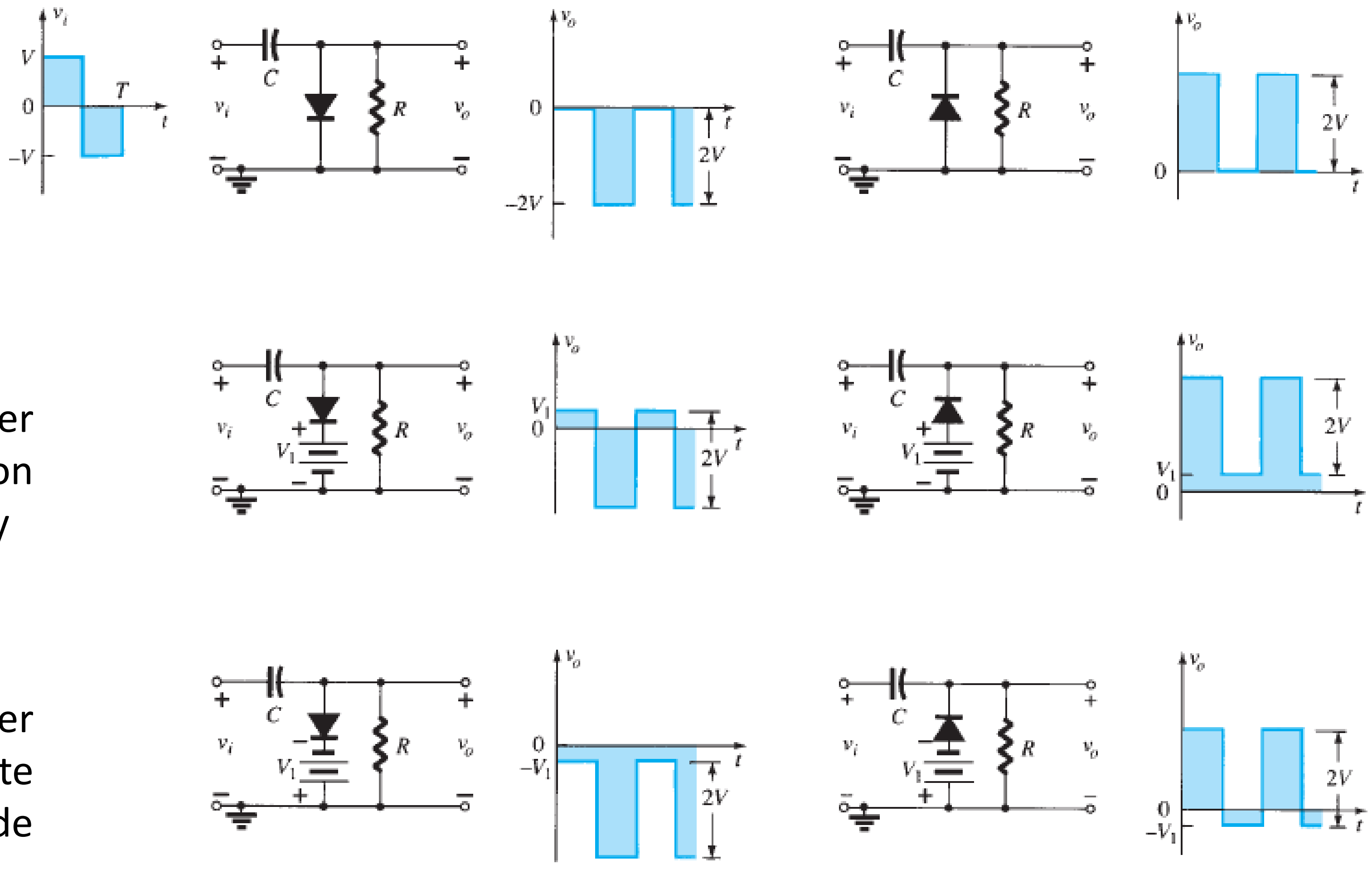


FIG. 2.100
Clamping circuits with ideal diodes ($5\tau = 5RC \gg T/2$).

References

- [1] R. Boylestad and L. Nashelsky, ‘Electronic Devices and Circuit Theory’, PHI, 7e, 2001.